STATEMENT OF WORK
National Oceanic and Atmospheric Administration

Project Title: HVAC Upgrade and BAS Controls Upgrade
Location: 4301 Rickenbacker Causeway Miami, FL 33149
Date: July 2026

1 Background:

The Atlantic Oceanographic and Meteorological Laboratory (AOML), located on Virginia Key, Florida is a research facility owned by the National Oceanic and Atmospheric Administration (NOAA). The main building was commissioned in 1973 and consists of roughly 78,000 square feet of office, laboratory, workshop, machinery, and storage space. There is also a 3,000 square foot warehouse and boat storage area behind the main building.

2 Purpose/Objective:

Install new HVAC Chiller and upgrades to the current HVAC systems, including the Metasys Building Management System. The result of this project is to allow building management to have climate control over individual spaces while maintaining code required air exchange rates. This includes all Metasys System devices and graphics to be operating properly. Also to provide a clean duct system throughout. Any equipment types and sizes and repairs allowed to be determined by the contractor must be approved by the Contracting Officer's Representative (COR) and or AOML Facilities Management.

There will be an allowable laydown area at the risk of the contractor. Advanced notice of work areas must be given at least two (2) days prior. The contractor is responsible for all dust and debris control as required by management. Contractor shall be responsible for the repair or refinish of any areas that are damaged during normal scope of work, including any damage to the surrounding work area utilized to gain access to the working area.

If the contractor identifies any ACM (Asbestos Containing Materials) during the course of work, the contractor must immediately notify AOML Management. The contractor must contain and handle such materials in compliance with all related Federal, State and Local rules and regulations.

3 Project Scope:

- Contractor shall provide ALL labor, materials and supervision necessary to complete the project.

- Below work will be completed onboard AOML.
- Contractor shall comply with submittal requirements detailed below.
- Contractor shall dispose of all demoed materials.
- Contractor shall coordinate with AOML staff for staging of materials and working hours. -
- Contractor shall follow all ASHRAE, ASTM, ICC, IECC, NFPA rules and regulations. -
- Contractor shall follow and comply with all NFPA 70, National Electrical Codes(NEC) -
- Contractor shall follow all city, state and federal safety rules and regulations
- Contractor shall confirm the quantity count and scope has been reviewed and verified to ensure project requirements are fulfilled.
- Contractor shall create design drawings/ Asbuilt drawings.

HVAC Chiller 290 Ton Daikin or like kind:

- New 290 Ton Chiller - with new dual compressors, all electrical, new valves, new controller, new display board, new sensors, connected to the Metasys system or other system to send text or email to AOML Facility if off. Complete hook up to CHW and CW water lines with all new connections. Proper insulation on the unit and any valves, water lines, etc. Replace to like kind, equal to existing or greater capacity.

HVAC Chill Water and Condensing Water lines/pipes located in the chiller room.

- Including all new connections and any sensors, gauges, separators, air vents, valves, meters, chemical line ports. Properly chemically treat all new makeup water at startup. Reinsulate all CHW lines.
- Connect devices to BAS and program accordingly, including any wiring, connectors, etc needed.

HVAC Expansion Tank

- Remove/Repair/Replace Expansion Tank in the Chiller Room area and replumb make up water for the CHW lines. This tank needs to be evaluated to see if it is needed. (Contractor will decide)

HVAC FCU4 in ceiling of the Chiller Room

- Replace and reinsulate all CHW lines.

Chiller Room Replace Refrigerant Detector/Evacuation Unit

- Replace, Including horn/strobe alarm and evacuation fan.
- Connect to Metasys System or other system to send alarm to AOML Facility via text and email.

Opticool Install Refrigerant Detector/Evacuation Unit

- Install appropriate size and type (determined by contractor) Including horn/strobe alarm and evacuation fan and any electrical, piping and wiring

needed.
- Connect to Metasys System or other system to send alarm to AOML Facility via text and email.

HVAC DOAS Unit
- Repair unit (determined by the contractor)
- Add reheat unit (type determined by the contractor)

HVAC 1st Floor Conference Room(Split System)
- Replace including any plumbing, electrical, supports, drainage, supplies needed.
- Type and size (determined by contractor)
- Replace any necessary ductwork
- Insulate all lines

HVAC Basement Unit
- Replace all duct work with sheetmetal. Including hangers, diffusers and insulate.
- Test and Balance unit
- Upgrade all unit controls to BACnet and program to the BAS. Including any software modifications to include sensor outputs and visual representation on BAS.

HVAC Duct Work
- Clean approximately 30 ft of all duct work extending from AHU-5, AHU-6, AHU-7, AHU-8 units.
- Inspect all duct work for leaks and unsecured insulation. Repair leaks and insulate as necessary.

HVAC VAV Heaters:
- Install thirty one (31) VAV electric heaters (type and size determined by the contractor) on AHU-5, AHU-6, AHU-7, AHU-8 including all required electrical/mechanical work.
- Reinsulate all VAV and ducting.
- Connect all VAV heaters to the BAS Meta System. Including any software modifications to include heater controls and visual representation on BAS.
- Install any new Siemens duct detectors that are necessary for heater installation and tie to VFDs and Siemens Fire Alarm System. Program and test.
- Ensure all other existing VAV Heaters on the 1st floor(AHU1) and 3rd floor (AHU11) are operating properly and make repairs or replacements as needed. This includes all Metasys System testing and any electrical or mechanical testing or control repair. Heater coils should be cleaned and ready for operation. Including any software modifications to include sensor

outputs and visual representation on BAS.

HVAC Humidity Sensors and CO2 Sensors:

- Install a humidity sensor on each floor and one outside.
- Install CO2 sensor on AHU1 return and connect it to FAU1 new BACnet controller.
- Program to control the louvers for outside air and connect and program to the BAS.
- Connect all humidity sensors to the BAS Meta System. Including any software modifications to include sensor outputs and visual representation on BAS.

BAS (Metasys System) Control Points:

- Connect Chiller to BAS system. Build BAS control points for chiller Start/Stop, Chill water pump start /stop, Condensing water start /stop. Data points of compressor operation status.
- Add data points for the opti-cool system.
- Add data points for Exhaust fan and operational controls.
- Including any software modifications to include sensor outputs and visual representation on BAS.

BAS (Metasys System) Controller Replacement and Upgrades:

- Install new BAS NOAA 1 and NOAA 2 BAS controller modules.
- Ensure communication to the BAS system.
- Including any software modifications to include sensor outputs and visual representation on BAS.
- Include any necessary devices and points to be added for the new Chiller.
- Newly added UPS Points, Opticool Points, and alarms, 1st floor upgrade of all N2 devices to Bacnet and reprogram any existing Backnet capable devices that were programmed for N2 back to Bacnet. Replace any defective control device (such as FAU1 Controller) and associated programming.
- Upgrading all CO2 Sensors, Humidity Control devices, Temperature Gauges and any associated programming
- Test all devices building wide to ensure all devices are communicating and operating properly. Including any software modifications to include sensor outputs and visual representation on BAS.
- Including software upgrades to the newest version.
- All electrical, piping, hangers and other associated materials must meet code requirements.
- Devices to be upgraded but not limited to:

Upgrade		**Change the N2 devices to BACnet and program**
Grd Floor	NAE Engine	Change to BACnet/Create Database/Program
Grd Floor	AHU Basement	Program to BACnet
1st Floor	FAU1 Controller	Change to BACnet/Program
1st Floor	AHU1 Controller	Program to BACnet
1st Floor	AHU1 Devices	Program to BACnet
1st Floor	AHU2 Controller	Program to BACnet
1st Floor	AHU2 Devices	Program to BACnet
1st Floor	Supply Air Sensors	Install and Program
4th Floor	EF 15A, 15B Controller	Program to BACnet
4th Floor	EF 9A, 9B Controller	Program to BACnet
Graphics	Redo	Graphics and user view
Communication Trunk	Trace	Trace
Test System	After Completed	Test Metasys System
Misc		Electrical, piping and wiring

Test and Balance:

- Conduct Test and balance of entire building HVAC system.
- Measuring all AHU, FCU, FAU, DOAS performance and all diffuser CFM readings. Included are any necessary corrections on inline physical dampers and all control and or programming of any control devices
- Modify BAS system, VAV, and Heaters to optimize building settings.

4 Period of Performance: Not to exceed 365 days.

5 Contract Requirements

5.1 Post Award/Preconstruction conference (Either virtual/in person)

The purpose of this meeting is to discuss and develop a mutual understanding of the administrative requirements of the Contract including but not limited to: daily reporting, invoicing, safety, site access, outage requests, schedule requirements, quality control, and schedule of prices, shop drawings, submittals, prosecution of the work, government acceptance, final inspections and contract close-out. The contractor must present and discuss their basic approach to scheduling the construction work and any required phasing. Contractor attendees must include the Project Manager, Superintendent, Site Safety and Health Officer (SSHO), Quality Control Manager (QCM) and major subcontractors. Superintendent, SSHO and QCM shall be designated in writing.

5.2 Safety and Health Plan: Site activities performed in conjunction with this project may pose safety hazards which require specialized expertise to effectively address and eliminate. The contractor shall be responsible for preparing and implementing an effective safety and health program, to include a site specific safety and health plan. The contractor's Safety and Health Plan shall conform to the requirements defined in U.S. Army Corps of Engineers (USACE) Safety Manual EM385-1-1. Plan must be submitted within 30 days of award and must be approved before the contractor is given notice to proceed with construction.

5.3 Quality Control (QC) Plan: The contractor shall develop, implement, and document an effective project specific quality control plan in accordance with USACE Engineer Regulation (ER) 1180-1-6, Construction Quality Management. This quality control plan shall be submitted to the COR for approval within 30 calendar days after contract award. All projects (over $750K) require a QC manager with a certificate of completion from a USACE Construction Quality Management for Contractors (CQM-C), 16 hour course within the last 5 years. This class is available online.

5.4 Cost Estimates: Contractor's proposals and cost estimates developed during the course of this contract, including modifications, shall follow UNIFORMAT II classification

5.5 Construction Status Reports: The contractor shall prepare daily construction status reports as shown in (ER) 1180-1-6, Construction Quality Management. Other daily report formats, with COR approval, are acceptable. At a minimum, these construction daily reports shall be submitted to the COR weekly. Bi-weekly project status meetings will be held.

5.6 Project Schedule: The contractor shall submit a detailed schedule showing all phases of design and construction, including testing, inspection/acceptance,

commissioning, training, and other activities as required. The schedule shall be submitted electronically in Microsoft Project compatible formats and .pdf format. Schedule updates are required to be submitted with monthly pay requests. For detailed scheduling requirements, see Exhibit 1 - Project Construction Progress Schedule.

5.7 Hours of Operation: The contractor is responsible for performing between the hours of 7:00 AM to 4:30 PM, Monday through Friday, except Federal holidays or when the Government facility is closed due to local or national emergencies, administrative closings, or similar Government directed facility closings. The contractor must at all times maintain an adequate work force for the uninterrupted performance of all tasks defined within this Statement of Work when the Government facility is not closed for the above reasons.

Work outside regular working hours requires COR approval. Request must be submitted 14 calendar days prior to such work to allow arrangements to be made by the Government for inspecting the work in progress, giving the specific dates, hours, location, type of work to be performed, contract number and project title. Based on the justification provided, the COR may approve work outside regular hours. See Section 10 Utilities for outage information.Contractor must at all times have someone on property that speaks fluent English.

5.8 Security:

5.8.1 Contractor Employees: All personnel used to perform work under this Contract shall be legal citizens of the USA, or legal immigrants to the USA with valid IDs. Prior to commencing work under this contract, the contractor shall submit or have their employee submit the required security forms as informed by the COR. The contractor, when notified that the Government rejected the suitability assessment forms shall either have the rejected forms made compliant and resubmitted or withdraw the employee from consideration from working under this contract. Contractor personnel may be required to obtain and wear badges in the performance of this task while on-site at a NOAA facility.

Provide a list of contract personnel of the contractor and subcontractors, including addresses and telephone numbers for use in the event of an emergency. As changes occur and additional information becomes available, update the information contained in previous lists.

5.8.2 Physical Security: The contractor shall be responsible for safeguarding all

Government property provided for contractor use. At the close of each work period, Government facilities, equipment, and materials shall be secured. Personnel may be subject to inspection of their personal effects when entering and leaving the facility. In addition, unscheduled inspections of personnel may be made when entering or while on site.

5.9 Environmental Protection

5.9.1. Provide and maintain, during the life of the contract, environmental protection as required. Plan for and provide environmental protective measures to control pollution that develops during construction. Plan for and provide environmental protective measures required to correct conditions that develop during the construction of permanent or temporary environmental features associated with the project. Protect the environmental resources within the project boundaries and those affected outside the limits of permanent work during the entire duration of this Contract. Comply with federal, state, and local regulations pertaining to the environment, including water, air, solid waste, hazardous waste and substances, oily substances, and noise pollution.

5.9.2 Tests and procedures assessing whether construction operations comply with Applicable Environmental Laws may be required. Analytical work must be performed by qualified laboratories; where required by law, the laboratories must be certified.

5.9.3 Prior to start of any onsite construction activities, perform a Preconstruction Survey of the project site with the Contracting Officer or the COR and the designated NOAA Environmental Representative. Survey shall include photographs showing existing environmental conditions in and adjacent to the site.

5.9.4 Minimize interference with, disturbance to, and damage to fish, wildlife, and plants, including their habitats. The protection of rare, threatened, and endangered animal and plant species identified, including their habitats, is the contractor's responsibility. Preserve the natural resources within the project boundaries and outside the limits of permanent work. Restore to an equivalent or improved condition upon completion of work. Confine construction activities to within the limits of the work indicated or specified.

5.9.5 If asbestos, lead paint, petroleum-contaminated soil, or suspected hazardous waste is found during construction that was not identified in the Contract documents, immediately notify the Contracting Officer and COR. Do not disturb this material until authorized by the Contracting Officer.

5.9.6. The contractor shall provide the necessary licensed Industrial Hygienist to confirm in writing the presence and scope of the hazardous material. The Contracting Officer will issue direction on how to handle the unexpected hazards described by the contractor.

5.9.7 Do not dispose of hazardous waste on Government property. Do not discharge any waste to a sanitary sewer, storm drain, or to surface waters or conduct waste treatment or disposal on Government property without written approval of the Contracting Officer.

5.9.8 The contractor shall minimize the presence of dust at all times, including during nonworking periods. Sprinkle or treat, with dust suppressants, the soil at the site, haul roads, and other areas disturbed by operations. Dry power brooming will not be permitted. Instead, use vacuuming, wet mopping, wet sweeping, or wet power brooming. Only wet cutting will be permitted for cutting concrete blocks, concrete, and bituminous concrete. Do not unnecessarily shake bags of cement, concrete mortar, or plaster. Since these products contain Crystalline Silica, comply with the applicable OSHA standard, 29 CFR 1910.1053 or 29 CFR 1926.1153 for controlling exposure to Crystalline Silica Dust.

5.9.9 The contractor shall implement best management practices (BMPs) for dust control and reducing equipment emissions during construction, including:

a) Watering exposed surfaces.
b) Covering haul trucks transporting loose material.
c) Removing visible mud or dirt track-out onto adjacent public roads.
d) Limit vehicle speeds on unpaved areas.
e) Minimize idling times.

5.9.10 Burning of trash or rubbish is not permitted on project site.

5.10 Monthly Invoices: Contractor is required to submit monthly invoices to COR via the IPP system. Invoices are due by the 7th of each month. The COR will review the invoices and either return them for corrections or process them for payment.

5.11 Site Restrictions: Contractor employees must park privately owned vehicles in an area designated by the COR. Employee parking must not interfere with existing and established parking requirements of the Government installation. Smoking is prohibited within and outside of all buildings on site, except in designated smoking areas. This applies to existing buildings, buildings under construction and buildings under renovation. Discarding tobacco materials other than into designated tobacco receptacles is considered littering and is subject to fines. The COR will identify designated smoking areas.

5.12 Dumpsters: Contractor must provide their own dumpster for all disposal of all trash

6 Inspection, Acceptance and Close Out

6.1 Periodic Inspections: In accordance with FAR 52.246-12 Inspection of Construction**,** the contractor shall maintain an adequate inspection system and perform such inspections as will ensure that the work performed under the contract conforms to contract requirements. The contractor shall maintain complete inspection records and make them available to the Government. All work shall be conducted under the general direction of the COR and is subject to Government inspection and test at all places and at all reasonable times before acceptance to ensure strict compliance with the terms of the contract

6.2 Commissioning: The contractor shall follow the commissioning procedures outlined in Exhibit 2 - Section 01 91 00, Total Building Commissioning. To make certain that the system functions properly in both heating and cooling seasons, the systems shall be commissioned a second time, in the season opposite of the initial commissioning, and before the end of the warranty period. The commissioning should demonstrate adequacy in weather representing both cooling and heating peak periods.

6.3 Final Inspection: The contractor shall provide no less than 10 working day notification to the COR prior to final inspection. The purpose of the final inspection is to ensure all systems are fully operational and ready for acceptance by the Government. If any system fails, the contractor shall be liable to pay any costs for government personnel to travel for re-inspection. The Site QC Manager shall sign off on request for final inspection, verifying all items have been completed and are operational and ready for Government acceptance.

6.4 BUILDER Inventory: NOAA utilizes the BUILDER Sustainment Management System (SMS) to maintain an inventory of all NOAA-owned assets, ensure visibility over equipment life cycles, and predict replacement time windows for major building systems. To assist the Government in the sustainment of BUILDER data, the contractor shall provide a complete list of all new equipment installed under this contract in the format provided in Exhibit 3 - BUILDER Inventory Update Form. The contractor shall also provide photographs of all newly installed equipment to the COR for use in the BUILDER inventory. Photographs shall include at least one photo containing a legible view of the nameplate data, if applicable. The completed BUILDER Inventory Update Form and photos must be submitted and accepted by the Government prior to final acceptance.

6.5 Operations and Maintenance (O&M): 30 days before the final inspection for the project, the contractor shall deliver to the COR an O&M manual for the contract which shall contain all instruction manuals and special directions provided by equipment manufacturers or vendors and all information necessary and appropriate for the start-up, operation, maintenance and repair of the facility, including complete equipment and system instructions and procedures for the start-up, operation, maintenance and repair of the facility. The contractor shall provide any specialized tools or equipment required to conduct O&M procedures as described in the O&M manual

6.6 Training: The contractor shall provide four (4) hours of training to the facility maintenance staff prior to turnover. As part of the training, the contractor shall inform the Government of any maintenance activities required to keep the warranty in effect. Contractor shall provide a minimum of 10 working days' notice and detailed agenda for training. Provide sign in sheet after completion.

6.7 As-Built Drawings: The contractor shall maintain an accurate set of as-built drawings to be updated daily and subject to inspection prior to approving monthly invoices. If the as-built drawings are not updated, then the invoice will be disapproved. At the completion of the contract, as-built drawings shall be submitted electronically in both .pdf and .dwg file formats and in hard copy to the COR.

6.8 Warranty Plan: At least 30 days before the final inspection, submit two sets of the warranty plan organized in a 3-ring binder. Include within the warranty plan all required actions and documents to assure that the Government receives all warranties to which it is entitled. The plan narrative must contain sufficient detail to render it suitable for use by future maintenance and repair personnel, who may not be familiar with this contract. The construction warranty period must begin on the date of project acceptance and continue for the full product warranty period. Conduct a joint six (6)-month and 11-month warranty inspection, measured from time of acceptance; with the contractor, COR and the owner representative.

6.8.1. The warranty management plan must include, but is not limited to, the following:

a) Roles and responsibilities of personnel associated with the warranty process, including points of contact and telephone numbers within the organizations of the contractors, subcontractors, manufacturers, or suppliers involved.
b) For each warranty, the name, address, telephone number, and e-mail of each of the equipment representatives nearest to the project location.

c) A list and status of delivery of Certificates of Warranty for extended warranty items, including roofs, HVAC balancing, pumps, motors, transformers, and for
commissioned systems, such as fire protection and alarm systems, sprinkler systems, and lightning protection systems.

d) As-Built Record of Equipment and Materials list for each warranted equipment, item, feature of construction or system indicating:

e) Name of item.

(1) Model and serial numbers.
(2) Location where installed.
(3) Name and phone numbers of manufacturers or suppliers.
(4) Names, addresses and telephone numbers of sources of spare parts.
(5) Warranties and terms of warranty. Include one-year overall warranty of construction, including the starting date of warranty of construction. Items which have warranties longer than one year must be indicated with separate warranty expiration dates.
(6) Cross-reference to warranty certificates as applicable.
(7) Starting point and duration of warranty period.
(8) Summary of maintenance procedures required to continue the warranty in force. (10) Cross-reference to specific pertinent Operation and Maintenance manuals. (11) Organization, names, and phone numbers of persons to call for warranty service.
(12) Typical response time and repair time expected for various warranted equipment.

f) The plans for attendance at the 6 and 11 month post-construction warranty inspections conducted by the Government.

6.9 Final Clean Up: Provide final cleaning in accordance with ASTM E1971. Remove debris from roadways and sweep/clean paved surfaces. Remove spoils from the project site. Repair all construction areas to previous condition. Remove waste and surplus materials, rubbish, and construction facilities from the site.

7 Submittals

The contractor shall provide the following submittals for Government review and approval before any onsite work can begin. Submittals shall be transmitted electronically and must be compatible with Microsoft Office applications, unless

otherwise approved by the COR. All construction drawing submittals shall also include a hard copy. Upon approval of submittals, the COR will notify the contractor.

Required Submittals prior to on-site work:

a) List of Contract Personnel

b) Design/Construction Project Schedule

c) Site Safety and Health Plan

d) Quality Control Plan

e) Appointment letters for site superintendent, QC manager, and SSHO f) Commissioning Plan

g) Material and Equipment Submittals

h) BUILDER Inventory Update Form

i) Training plan

j) Warranty plan

k) As-built drawings

l) Operation and Maintenance Manuals

8 Government Furnished Equipment (GFE): The government will provide the following equipment for this contract.

None

9 Utilities:

9.1 Utility Outage Requests and Utility Connection Requests: Schedule work to minimize outages. For utility outages and connections required during the execution of work that affect existing systems, schedule outside the regular working hours or on weekends, as approved by theCOR. Coordination with the government for all outages is required to ensure there is no impact to existing research. No additional payment will be provided for utility outages and connections required to be performed outside the regular work hours. Submit requests for utility outages and connections in writing to the COR for approval at least 21 calendar days in advance of the time required. In each request, state the system involved, area involved, approximate duration of outage, and the nature of work involved.

9.2 Temporary Utilities: Provide temporary utilities required for construction. Materials may be new or used, must be adequate for the required usage, not create unsafe conditions, and not violate applicable codes and standards.

9.3 Electrical: The Government will make reasonable amounts of electricity available to the contractor for the duration of the contract. Access to existing power must be coordinated with the Government. All temporary electrical connections must be in accordance with all applicable codes and station policies. Connections to the existing power system will be at contractor's expense.

9.4 Water: Potable water connections to existing facilities to provide water for construction purposes will be made available. Connections will need to be coordinated with site personnel and will be at the contractor's expense.

9.5 Sanitation: Provide and maintain within the construction area minimum field-type sanitary facilities in accordance with EM 385-1-1 Section 02. Locate the facilities behind the construction fence or out of the public view. Clean units and empty wastes at least once a week. Government toilet facilities will not be available to contractor personnel.

9.6 Data/Telephone: No government provided telephone or data is available at this site. Make arrangements and pay all costs for services as desired.

9.7 Fire Protection: Provide temporary fire protection equipment for the protection of personnel and property during construction. Remove debris and flammable materials daily to minimize potential hazards.

10 Exhibits

Exhibit 1 - Project Construction Progress Schedule
Exhibit 2 - Section 019100 Total Building Commissioning
Exhibit 3 - BUILDER Inventory Update Form

11 References

Work done under this contract shall utilize the latest issue of the publication dated at the time of award. When a required publication is not referenced in this list, the contractor shall utilize one that has national applications. Where conflicts arise between publications, the most stringent shall apply.

UNIFIED FACILITY CRITERIA

UFC 1-200-01 DoD Building Code (General Building Requirements), With Change 2

US ARMY CORPS OF ENGINEERS

EM-385-1-1 Safety Manual

ER 1180-1-6, Construction Quality Management

CODE OF FEDERAL REGULATIONS (CFR)

29 CFR 1910 Occupational Safety and Health Standards - General Construction 29 CFR 1926 Occupational Safety and Health Standards - Construction Industry 40 CFR 61 National Emission Standards for Hazardous Air Pollutants

FEDERAL STANDARDS (FED STD)

FED STD 313A Material Safety Data Sheets, Preparation and the Submission FED STD 795 Uniform Federal Accessibility Standards